The Lazaros–Eudora Method (LEM): A Topological Framework for User–LLM Interaction Dynamics

Lazaros Varvatis
Independent researcher

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This draft adds the Toy V2 robustness check and Amendment 1 of 8 November 2026.
The published v2.2.0 and archived v3/v4 materials remain unchanged.

Abstract

We present two synthetic pipeline checks for the Lazaros–Eudora Method (LEM), a conceptual framework that treats user–LLM interactions as trajectories in latent representation space. The first demonstrates geometric identifiability of user-conditioned trajectories in a controlled synthetic dynamical system, with perfect nearest-neighbor re-identification across $N = 500$ users and ten independent seeds, compared to a random baseline of 0.002. The second extends an archived sink/cycle example to a pre-specified multi-seed check with matched controls and a fixed joint geometric predictor.

For this toy and these fixed predictors, Amendment 1 assigns the overall category *incremental value*: in the primary comparison against GAUSS, $\mathrm{AUC}(H_1) = 0.994$ versus 0.828 for the joint geometric predictor, with a paired increment of 0.166 (95% interval $[0.109, 0.235]$) when adding H_1. The original rule assigned *partial*.

Neither check establishes user signatures in language-model activations or performance in real-user populations. Geometric and topological summaries remain candidate measurements for empirical work; LEM remains a research program requiring independent real-model tests.

1 Introduction

Current large language model evaluation remains dominated by static, turn-local metrics. These metrics can assess answer quality, benchmark performance, or stylistic alignment, but they do not directly describe the temporal structure of repeated user–model interaction. As a result, an important object remains under-modeled: the trajectory induced by a specific user interacting with a model across multiple turns.

The Lazaros–Eudora Method (LEM) is a framework designed to study this object. Its central premise is that user–LLM interaction should be modeled as a dynamical trajectory in latent space rather than as a collection of isolated prompt embeddings or output tokens. In this view, repeated interaction may generate structured regimes—for example convergence, recurrence, or instability—that are more naturally described in the language of dynamical systems and topology than in the language of static vector similarity alone.

This paper makes a limited but concrete contribution. It does not claim to establish a full theory of human cognition or a validated description of real frontier-model internals. Instead, it contributes:

1. a compact conceptual formalization of LEM;
2. a distinction between geometric and topological notions of interaction signatures;
3. two toy-model pipeline demonstrations showing how constructed user vectors and trajectory classes can be measured in controlled synthetic settings, including a pre-specified multi-seed Toy V2 check with matched controls and a dated amendment evaluating a fixed joint geometric predictor.

The intended contribution is therefore methodological: to provide a testable framework for studying user–LLM interaction dynamics and to identify which parts of the theory are already operational, which remain open, and which are presently speculative.

2 Conceptual Framework

2.1 Interaction trajectories

We define an interaction trajectory as a time-ordered sequence of states

$$x_1, x_2, \ldots, x_T \in \mathcal{X},$$

where $\mathcal{X}$ is a latent or constructed representation space associated with a multi-turn user–LLM interaction. In practice, $\mathcal{X}$ may refer to internal activations, reduced coordinates, or synthetic latent states in a toy model. For inference with frozen weights, the activation at a turn depends on the supplied context. Returning to the same weights with empty history does not retain an earlier dialogue without explicit memory or training. Context-carried signatures must be distinguished from changes stored in model parameters.

This shift from isolated states to trajectories is the first core move of LEM. The object of study is not merely the embedding of a prompt or response, but the path induced by repeated interaction.

2.2 Cognitive signatures

A *cognitive signature* is a compact representation derived from an interaction trajectory that preserves salient structural properties of user–model dynamics. In the present paper, we distinguish two operational levels:

1. **Geometric signatures**, defined through distances, directional similarity, and nearest-neighbor re-identification in a controlled latent system;
2. **Topological signatures**, defined through persistent homology, especially the presence or absence of robust H_1 features.

This distinction is a proposed measurement strategy. A topological summary captures loops in a point cloud, but the present examples do not show that it outperforms geometric summaries of the same trajectories.

2.3 Attractor-like interaction regimes

LEM uses the term *attractor* in a modeling sense. The claim is not that real human cognition has been recovered from LLM traces, but that repeated interaction can be approximated as a dynamical process whose trajectories may approach stable regimes. In the minimal setting considered here, these include:

- **Convergent regimes**, approximated by a stable fixed point;
- **Cyclic regimes**, approximated by a stable limit cycle;
- **Unstable or noisy regimes**, in which no compact structural summary is stable.

2.4 Core hypotheses

We state four hypotheses. The synthetic examples instantiate aspects of the first three by construction; none is empirically established for real user–LLM interaction by these examples.

H1: Geometric identifiability. User-conditioned interaction dynamics can generate trajectories that are re-identifiable above chance in a controlled latent system.

H2: Regime differentiation. Convergent and cyclic interaction regimes generate structurally distinct trajectory classes.

H3: Topological utility. Persistent homology captures structural differences between such regimes, especially through the presence or absence of persistent H_1 features.

H4: Cross-version stability. User-specific interaction signatures may remain partially stable across related model variants or model families. In this paper, H4 is treated as a theoretical target rather than an empirically established claim.

3 From Geometric Similarity to Topological Structure

Many current approaches to personalization, user modeling, and dialogue adaptation rely on static similarity measures, clustering, or compressed user embeddings. These approaches are useful, but they typically summarize multi-turn interaction as a point estimate or a low-order statistic.

LEM instead treats interaction as a trajectory and therefore distinguishes between two levels of structure:

1. **Geometric structure**: whether trajectories lie near each other or near user-conditioned attractors;
2. **Topological structure**: whether trajectories exhibit recurrence, loop-like organization, or topological triviality.

This motivates the use of topological data analysis (TDA), especially persistent homology. In the current work, the key topological observable is persistent H_1 structure, which can detect a sampled loop in a constructed point cloud. A loop alone does not establish a dynamical attractor: recurrence and return after perturbation require temporal and interventional evidence.

We explicitly do *not* claim that topological features directly reveal psychological essence or consciousness. A claim of added value over geometry requires a matched nonloop control and an independent comparison with geometric summary statistics; neither is included in the original Toy V2 example.

4 Toy-Model Pipeline Checks

The present paper does not validate LEM on real model internals. These deliberately constructed examples check that the numerical pipeline behaves as expected under known dynamics. An implementation defect could make such a check fail. Success, however, does not provide a test of whether user-specific signatures or attractors exist in language models. That question requires independent data and a falsifiable real-model design.

We therefore present two toy checks with distinct roles. The first constructs well-separated user-indexed synthetic vectors and checks their geometric identifiability. The second starts with the archived sink/cycle example and adds a pre-specified multi-seed comparison with matched controls and held-out instances. Its amended joint geometric comparison is described below; the original V1 numerical files and original Toy V2 generator remain unchanged.

4.1 Toy Experiment 1: Geometric Identifiability

The first toy experiment is a geometric sanity check for the notion of user-specific interaction structure. We consider a latent dynamical system of the form

$$x_{t+1} = (1 - \alpha - \beta)\, x_t + \alpha\, s_i + \beta\, d_m + \varepsilon_t, \tag{1}$$

where $x_t \in \mathbb{R}^d$ denotes the latent state at time t, s_i is a user-specific signature vector, d_m is a domain or model attractor, and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2 I_d)$ is isotropic Gaussian noise. The interpretation is deliberately minimal: s_i models a user-conditioned directional bias in latent space, while d_m models task- or model-level regularization toward a shared region. This is a controlled synthetic separability regime and does not simulate a deployed user–model interaction.

4.1.1 Setup

We instantiate $N = 500$ users and $M = 2$ model domains in a $d = 512$-dimensional embedding space. User vectors and initial domain vectors are independently drawn from $\mathcal{N}(0, I_d)$ and ℓ_2-normalised. The second domain is then replaced by $d_1 = \text{normalize}(d_0 + 0.4z)$, with a fresh $z \sim \mathcal{N}(0, I_d)$; the two final domains are therefore not independent constructions. Trajectories use $\alpha = 0.4$, $\beta = 0.3$, $\sigma = 0.15$, and $T = 200$. Empirical signatures retain the published definition:

$$\hat{s}_i^{(m)} = \frac{1}{K} \sum_{j=0}^{K-1} x_{\tau-j}^{(i,m)} - \beta\, d_m, \qquad K = 40. \tag{2}$$

Here τ denotes the last stored state index, so exactly K states are averaged. Vectorized training performs T updates ($\tau = T$); scalar test trajectories store the initial state and $T - 1$ updates ($\tau = T - 1$). These original execution conventions are preserved. The deterministic fixed point of Eq. (1) is

$$x^* = \frac{\alpha\, s_i + \beta\, d_m}{\alpha + \beta}. \tag{3}$$

In the noise-free limit, the retained signature estimator is thus $(4/7)s_i+(9/70)d_m$. Subtracting βd_m does not fully remove the fixed point's domain component. Cross-domain cosine values describe this operational signature, including residual domain dependence, rather than a purely user-specific estimate.

The scaled experiment uses ten seeds: $42, 123, 999, 2024, 7, 314, 1337, 77, 256, 88$. The robustness experiment below instead uses one fixed seed.

4.1.2 Metrics

We evaluate three observables: (i) mean ℓ_2 distance between the empirical mean of the last K states and the theoretical fixed point of Eq. (3); (ii) nearest-neighbor re-identification accuracy over $N_{\text{test}} = 500$ fresh test queries per model; (iii) same-user versus different-user cosine similarity of empirical signatures across model domains.

4.1.3 Results

Table 1 reports summary statistics across all ten seeds. Standard deviations are sample standard deviations (ddof = 1), calculated from the six-decimal per-seed exports, as in the original Pandas aggregation. The corrected distance uses Eq. (3); the archival implementation mistakenly omitted its denominator. NN accuracies and cosine values are unchanged in a matched before/after run. The source and numerical provenance are recorded in `CORRECTION_NOTE.md` and the accompanying result manifest.

Table 1: Toy Experiment 1 — scaled results in the controlled synthetic separability regime (d=512, N=500, T=200, 10 seeds). Std denotes sample SD (ddof = 1).

Metric	**Mean**	**Std**	**Min**	**Max**
Random baseline accuracy	0.0020	—	—	—
NN re-id accuracy	1.0000	0.0000	1.0000	1.0000
Mean attractor distance	0.7597	0.0008	0.7583	0.7607
Cosine (same user)	0.3570	0.0020	0.3538	0.3609
Cosine (different user)	0.0022	0.0018	-0.0011	0.0056

Nearest-neighbor re-identification accuracy reaches 1.000 ± 0.000 across all seeds—a 500× improvement over the random baseline of 0.002 in this synthetic setting. Same-user cosine similarity (0.3570 ± 0.0020) is far above the near-zero cross-user similarity (0.0022 ± 0.0018), confirming geometric separability.

4.1.4 Robustness analysis

We evaluate the original grid

$$\sigma \in \{0.05, 0.15, 0.30, 0.50, 0.75, 1.00, 1.50, 2.00\}.$$

Dimensions, user/domain counts, weights, trajectory lengths and window lengths match the scaled experiment. V1b uses 300 fresh test queries per model (600 total), compared with 500 per model in the scaled experiment. Seed 42 generates the fixed user/domain setup and is reset to 42 before every noise level, preserving the original shared random sequence across levels. This is a single-seed comparison; no across-seed standard deviation is reported for V1b.

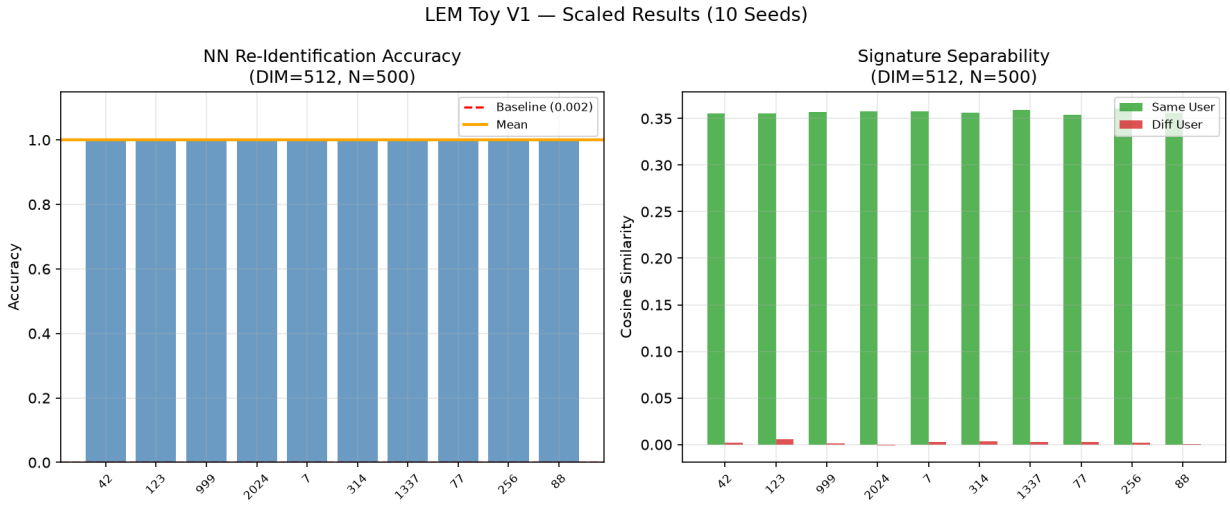


Figure 1: Toy Experiment 1. **Left:** NN re-identification accuracy across ten seeds (orange line = mean, red dashed = random baseline). **Right:** cosine similarity for same-user (green) versus different-user (red) signature pairs. All results are within the controlled synthetic separability regime described in Section 4.1.

Table 2: Robustness to noise level σ (d=512, N=500, seed = 42; 300 queries per model). Attr. Dist. is the corrected tail-mean distance.

σ	**NN Acc.**	**Attr. Dist.**	**Cos. (same)**	**Cos. (diff)**
0.05	1.0000	0.2533	0.8037	0.0045
0.15	1.0000	0.7599	0.3545	0.0019
0.30	0.2300	1.5198	0.1219	0.0006
0.50	0.0200	2.5329	0.0469	0.0002
0.75	0.0050	3.7994	0.0207	0.0001
1.00	0.0033	5.0659	0.0112	0.0000
1.50	0.0033	7.5988	0.0043	0.0000
2.00	0.0017	10.1317	0.0018	0.0000

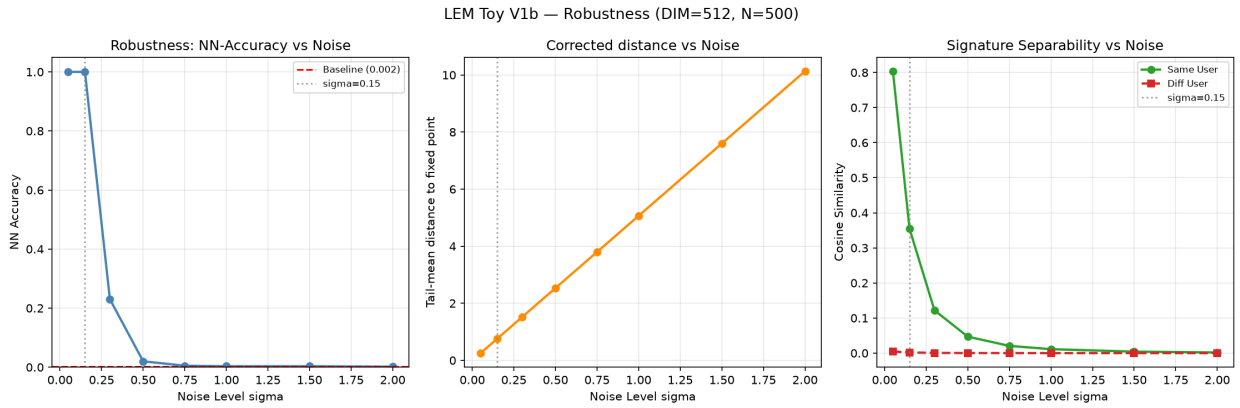


Figure 2: Robustness of LEM signatures on the sampled noise grid. **Left:** NN accuracy drops between $\sigma = 0.15$ and 0.30; no threshold at 0.25 was measured. **Centre:** corrected tail-mean distance to the fixed point. **Right:** decreasing same-user cosine similarity and near-zero different-user similarity.

Figure 2 shows perfect re-identification at the sampled levels $\sigma = 0.05$ and 0.15, followed by accuracy 0.2300 at 0.30. The grid has no observation at 0.25 and does not locate a precise transition. At 0.50, accuracy is 0.0200, ten times the random baseline 0.0020. At the larger sampled levels 0.75–2.00, observed accuracies range from 0.0050 to 0.0017; the single-seed query counts do not establish exact equality to chance. Same-user cosine similarity decreases with noise while different-user cosine remains near zero. These observations alone do not isolate a causal mechanism for classification errors.

The corrected tail-mean distance is approximately proportional to σ over the sampled grid. Its magnitude must be distinguished from fluctuations of an individual state. For $\rho = 1 - \alpha - \beta = 0.3$, the stationary variance of one state coordinate is $\sigma^2/(1-\rho^2)$. Averaging K consecutive, temporally correlated states gives per-coordinate variance

$$v_K = \frac{\sigma^2}{(1-\rho^2)K^2}\left[K + 2\sum_{h=1}^{K-1}(K-h)\rho^h\right].$$

Thus the stationary root-mean-square norm of the tail-mean error is $\sqrt{d\,v_K}$, approximately 5.069σ for $d = 512$, $K = 40$. The reported metric averages norms and is not exactly this RMS quantity; finite sampling also matters. The dependence on dimension, window length and temporal correlation precludes interpreting the observed slope as a universal single-state noise law.

Perfect re-identification here is a property of the high-dimensional synthetic construction, in which random user unit vectors are nearly orthogonal. It is not evidence of real-world user identifiability. The sampled loss of accuracy characterizes this particular simulation and does not establish a general operating threshold for LEM.

4.2 Toy Experiment 2: Topological Class Separation

The second toy experiment is an illustrative check of the persistent-homology pipeline on two prescribed dynamical systems. Its setup supplies the sink and cycle in advance; it does not test whether user interactions produce either.

4.2.1 Setup

We construct two synthetic trajectory classes:

1. a **convergent regime**, modeled as a stable sink in two dimensions:
$$\dot{x} = -\lambda x, \quad \dot{y} = -\lambda y, \quad \lambda = 0.5, \tag{4}$$
initialised at $(x_0, y_0) = (2.0,\ 1.5)$;
2. a **cyclic regime**, modeled as a stable limit cycle via the Van der Pol oscillator:
$$\dot{x} = y, \quad \dot{y} = \mu(1 - x^2)y - x, \quad \mu = 1.5, \tag{5}$$
initialised at $(x_0, y_0) = (0.1,\ 0.1)$.

Both systems are integrated over T=40 time units with Δt=0.1 (401 steps), then embedded into $\mathbb{R}^{16}$ via random linear projections $A \in \mathbb{R}^{2\times 16}$ with additive Gaussian noise (σ=0.05). The executed code samples a separate projection for each trajectory; their geometry and scale are therefore not controlled to match. Persistent homology in degree H_1 is computed via the Vietoris–Rips filtration using `ripser` [3, 14].

4.2.2 Metric

For each trajectory we compute the degree-1 persistence diagram $\mathrm{PD}(H_1)$. The key quantity is the *maximum persistence*

$$p_{\max} = \max_{(b,d)\,\in\,\mathrm{PD}(H_1)} (d - b),$$

where b and d denote the birth and death of a homological feature. A large $p_{\max}$ summarizes the longest detected one-dimensional feature in each sampled point cloud. The historical experiment calls the following quotient its "SNR"; it is a descriptive ratio of two maxima, not an independently calibrated signal-to-noise statistic:

$$\mathrm{SNR} = \frac{p_{\max}^{\mathrm{cyclic}}}{p_{\max}^{\mathrm{convergent}} + \epsilon}, \qquad \epsilon = 10^{-9}.$$

4.2.3 Results

Table 3: Toy Experiment 2 — one constructed sink/cycle pair (d=16, T=40, seed = 1337); descriptive results only.

Metric	**Convergent**	**Cyclic**
Max H_1 persistence	0.0326	5.2649
Mean H_1 persistence	0.0092	0.1972
Ratio (historically "SNR")	161.6	

The cyclic regime produces a maximum H_1 persistence of 5.265, compared to 0.033 for the convergent regime—a ratio of 161.6× for this pair. A threshold $p_{\max} > 0.1$ labels these two observed examples correctly; it was not selected and evaluated on separate samples, so no error rate is estimated. In that archived example there is one seed, no matched-scale noncyclic control, no held-out pairs and no comparison against simpler geometric descriptors. The large ratio therefore neither establishes that persistent homology reliably separates unseen regimes nor demonstrates that topology adds value for LEM's user-level hypotheses.

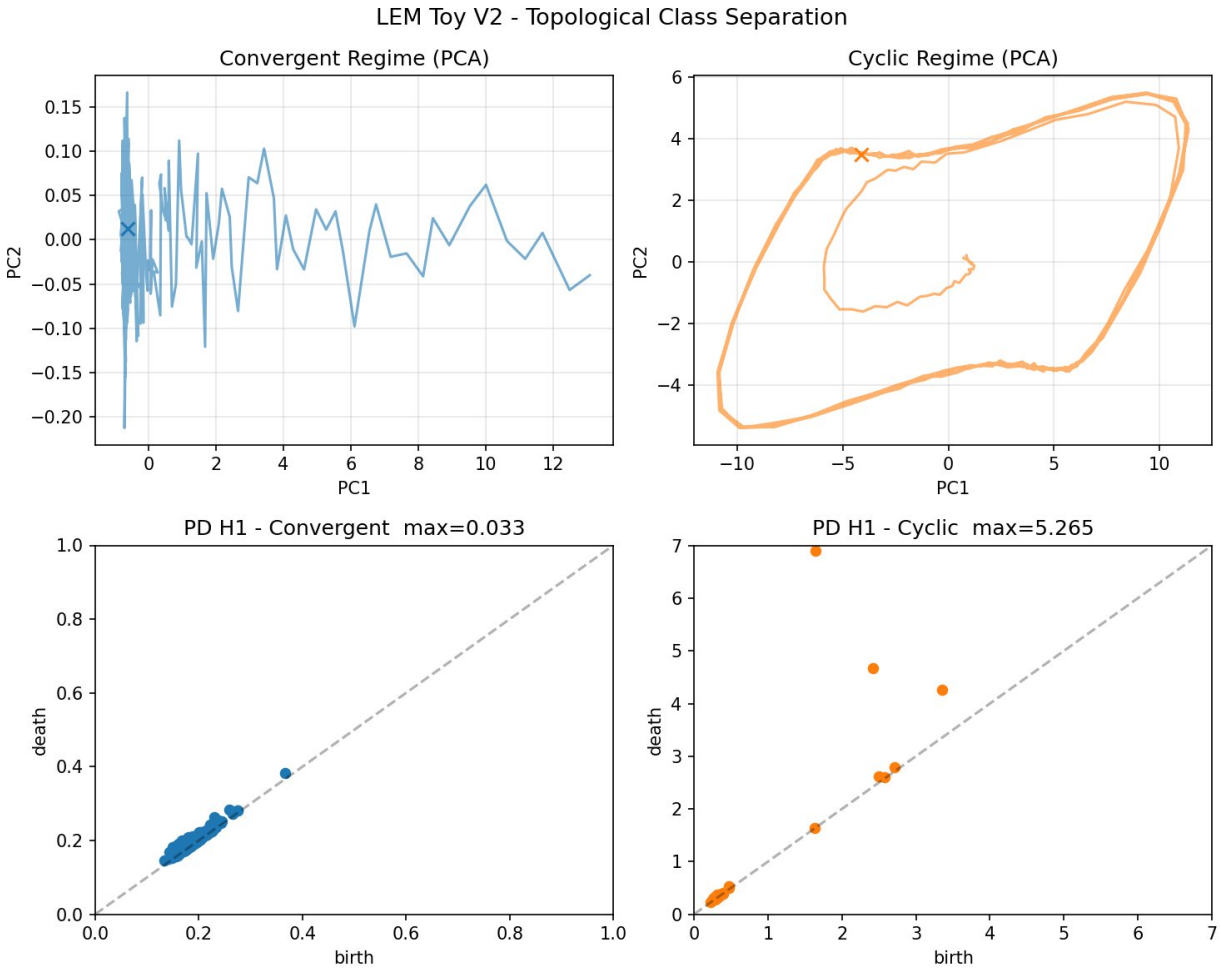


Figure 3: Toy Experiment 2: topological class separation. **Top:** PCA projections of the convergent (left, blue) and cyclic (right, orange) trajectories embedded in $\mathbb{R}^{16}$. **Bottom:** degree-1 persistence diagrams. The convergent diagram shows only diagonal noise ($p_{\max}$=0.033); the cyclic diagram contains a prominent off-diagonal generator ($p_{\max}$=5.265) for the constructed limit cycle. The historical ratio called "SNR" is 161.6 for this single pair.

4.3 Pre-specified multi-seed check of Toy V2

Protocol, amendment and chronology. The original design was committed as `c7d44d1` before any computation. Its outputs were committed as `ee90b9f`. Amendment 1, dated 8 November 2026, was drafted by reviewers who had not seen the results, then supplied following external review and committed as `cf66e01` before supplemental computation. Its purpose was to compare H_1 with a joint geometric predictor, add a uniform control, and explicitly test invariance to temporal shuffling. The original results had already been computed, inspected by the executing assistant and reported in the chat before the amendment arrived. Thus the amended analysis is not presented as preceding all access to the original results. Both frozen versions are reported; original output hashes remain unchanged. The exact specifications and seed assignments are in `PROTOCOL.md`.

Generators, controls and measurement window. We retain the original sink and Van der Pol generators, numerical integration, initial conditions, $\mu = 1.5$, $\lambda = 0.5$, $T = 40$ and $\Delta t = 0.1$. The window W consists of all 401 points entering persistence, including transients; the original has no burn-in removal or subsampling. Numerical helper functions were copied verbatim, with only dedentation, from `experiments/lem_simulations.py` at source commit `e2c0487` (lines 591–615), and checked against that source.

The amended name **GAUSS** denotes the original **DISK**: a stationary 2-D Gaussian OU process discretized as AR(1). Its common lag-1 coefficient is estimated by pooling the centered CYCLE coordinates on W. The start is drawn from the stationary distribution. The realized sample is then centered, whitened by its own sample covariance, colored by CYCLE's sample covariance and shifted to CYCLE's mean. **UNIFORM** starts with independent uniform points in an ellipse and undergoes the same exact finite-sample affine moment matching. The sample-dependent matching step couples the final points; uniformity and independence describe the initial draws. **LINE** is a finite sinusoid on CYCLE's first principal axis, with the dominant nonzero FFT frequency and exact sample mean and total-variance matching. Its amplitude uses the finite sinusoid's sample variance, not $A^2/2$.

A common unnormalized Gaussian 2×16 projection per seed is primary; independent projections per trajectory are secondary. Independent Gaussian observation noise is added after projection. Original random streams and per-seed statistics are preserved; UNIFORM uses a separate control stream and is appended after the original classes in the embedding stream. Before observation noise, the GAUSS/UNIFORM moment matching also matches g_1 and g_2 under a common projection; these serve as plausibility checks, not as the main individual geometric competitor.

Statistics, frozen predictors and uncertainty. The individual statistics are maximum finite Vietoris–Rips H_1 lifetime; g_1, the radius of gyration on all W; g_2, the largest divided by the second-largest PCA covariance eigenvalue; and g_3, the coefficient of variation of centroid distances in the first two principal components. We additionally report $g_{1,\mathrm{half}}$ on the second half for comparability; it does not enter categorization. For each statistic/comparison/noise/projection, sign and threshold are selected on calibration seeds by Youden's criterion and frozen.

The joint geometric predictor G is logistic regression on standardized (g_1, g_2, g_3); $G + H_1$ additionally includes H_1. Both use L2 regularization with $C = 1.0$ fixed. Scalers and models are fitted exclusively to calibration seeds, with no subsequent tuning. The control is the positive class. For G and $G + H_1$, accuracy uses the fixed probability cutoff 0.5; H_1 accuracy uses its original frozen threshold. At noise 0.05, seeds 0–99 calibrate and 100–199 evaluate, with all classes sharing each seed. At each other noise level $(0.01, 0.025, 0.1, 0.2, 0.5)$, seeds 0–19 calibrate and 20–39 evaluate. Levels and projection modes are never pooled. Percentile 95% bootstrap intervals use 1000 resam-

ples of whole evaluation seeds (fixed seed 2300), keeping all classes and predictions of each seed together. Models, signs and thresholds are not refitted in resamples; ΔAUC uses paired differences from those same resamples.

Table 4: Amendment 1 main evaluation: CYCLE versus each control, noise 0.05, common projection, 100 evaluation seeds per class. Entries show AUC and paired 95% bootstrap interval; ΔAUC is AUC($G + H_1$)−AUC(G).

Control	H_1	G	$G + H_1$	ΔAUC
GAUSS	0.9942 (0.9842, 1.0000)	0.8276 (0.7547, 0.8859)	0.9933 (0.9793, 1.0000)	0.1657 (0.1087, 0.2347)
UNIFORM	1.0000 (1.0000, 1.0000)	0.5745 (0.4824, 0.6677)	1.0000 (1.0000, 1.0000)	0.4255 (0.3323, 0.5176)
LINE	1.0000 (1.0000, 1.0000)	1.0000 (1.0000, 1.0000)	1.0000 (1.0000, 1.0000)	0.0000 (0.0000, 0.0000)

Table 5: Accuracy under the frozen H_1 threshold or fixed logistic probability cutoff 0.5, with 95% bootstrap intervals, on the same amended primary evaluation.

Control	H_1	G	$G + H_1$
GAUSS	0.9850 (0.9650, 1.0000)	0.7750 (0.7150, 0.8300)	0.9900 (0.9750, 1.0000)
UNIFORM	0.9950 (0.9850, 1.0000)	0.5950 (0.5250, 0.6650)	1.0000 (1.0000, 1.0000)
LINE	1.0000 (1.0000, 1.0000)	1.0000 (1.0000, 1.0000)	1.0000 (1.0000, 1.0000)

Amended decision rules and result. Categories are determined separately for GAUSS and UNIFORM on the primary evaluation. *clear advantage* requires AUC(H_1) ≥ 0.95 and AUC(G) ≤ 0.65. *incremental value* requires AUC(H_1) ≥ 0.95, AUC(G) > 0.65, ΔAUC≥ 0.05, and a strictly positive lower 95% bootstrap limit for ΔAUC. *tie* applies when AUC(H_1) ≥ 0.95 and neither preceding condition holds. *not robust* applies when AUC(H_1) < 0.95: H_1 alone does not recognize the constructed loop reliably, not "not significant." The overall category is the weaker of the two, ordered as these four labels above. LINE is a plausibility probe and does not enter categorization.

The amended primary result is **incremental value** for GAUSS and **clear advantage** for UNIFORM; the overall category is **incremental value**. Against GAUSS, H_1 AUC is 0.9942 $[0.9842, 1.0000]$, G AUC is 0.8276 $[0.7547, 0.8859]$, and $G + H_1$ AUC is 0.9933 $[0.9793, 1.0000]$; the paired increment is 0.1657 $[0.1087, 0.2347]$. Against UNIFORM the corresponding AUCs are 1.0000, 0.5745 and 1.0000, with increment 0.4255 $[0.3323, 0.5176]$. These categories apply only to the specified toy controls and fixed predictors; they do not establish an advantage over all geometric methods.

Original committed analysis, retained without replacement. The original evaluation compared individual statistics only. R1 is **H1 separates sink and cycle**, since AUC(H_1, Va) $= 1.0000$. Its R2, based solely on Vb (CYCLE versus DISK, now called GAUSS), is **partial**: H_1 AUC is 0.9942, g_1 is 0.5000, g_2 is 0.5003, and g_3 is 0.8321. The original rules classify all g AUCs ≤ 0.65 as *clear advantage*, a best g AUC strictly between 0.65 and 0.95 as *partial*, any g AUC ≥ 0.95 as *no clear*

Table 6: Amended primary evaluation: individual statistics, AUC and accuracy with 95% bootstrap intervals. The comparison is CYCLE versus the named control; $g_{1,\mathrm{half}}$ is supplementary and order-dependent.

Comparison / control	Statistic	AUC (95% CI)	Accuracy (95% CI)
GAUSS	H_1	0.9942 (0.9842, 1.0000)	0.9850 (0.9650, 1.0000)
GAUSS	g_1	0.5000 (0.4962, 0.5031)	0.5000 (0.5000, 0.5000)
GAUSS	g_2	0.5003 (0.4968, 0.5045)	0.5000 (0.5000, 0.5000)
GAUSS	g_3	0.8321 (0.7637, 0.8843)	0.7000 (0.6449, 0.7550)
GAUSS	$g_{1,\mathrm{half}}$	0.7920 (0.7473, 0.8369)	0.7100 (0.6599, 0.7600)
UNIFORM	H_1	1.0000 (1.0000, 1.0000)	0.9950 (0.9850, 1.0000)
UNIFORM	g_1	0.5011 (0.4983, 0.5053)	0.5000 (0.5000, 0.5000)
UNIFORM	g_2	0.4997 (0.4961, 0.5034)	0.5000 (0.5000, 0.5000)
UNIFORM	g_3	0.4794 (0.3921, 0.5634)	0.6150 (0.5650, 0.6700)
UNIFORM	$g_{1,\mathrm{half}}$	0.6478 (0.6308, 0.6755)	0.6150 (0.5750, 0.6600)
LINE	H_1	1.0000 (1.0000, 1.0000)	1.0000 (1.0000, 1.0000)
LINE	g_1	0.5129 (0.4678, 0.5590)	0.5250 (0.4800, 0.5700)
LINE	g_2	1.0000 (1.0000, 1.0000)	0.9950 (0.9850, 1.0000)
LINE	g_3	0.9200 (0.8600, 0.9700)	0.9500 (0.9150, 0.9750)
LINE	$g_{1,\mathrm{half}}$	0.6381 (0.5961, 0.6859)	0.6350 (0.5900, 0.6850)

advantage, conditional on H_1 AUC ≥ 0.95; otherwise they assign *not robust*. Vc (CYCLE versus LINE) remains outside R2; its g_2 AUC is 1.0000. The different original and amended category labels reflect different specified comparators and decision rules; one has not been substituted for the other.

Moment matching, shuffling, reproducibility and limits. The maximum GAUSS pre-projection mean and covariance residuals across all seeds are 3.29×10^{-15} and 2.66×10^{-15}; for UNIFORM they are 3.47×10^{-16} and 2.44×10^{-15}. LINE mean and total-variance residuals are 4.86×10^{-17} and 1.78×10^{-15}. The target pooled AR(1) coefficient is 0.99051344; realized empirical lag-1 estimates range from 0.95540 to 0.99667 before affine matching and from 0.94809 to 0.99575 after it. A fixed affine map commutes with a scalar AR(1) transition, but a sample-dependent map does not guarantee stationary independent Gaussian innovations or exact empirical lag-1 matching in the corrected ensemble.

All 800 CYCLE/CYCLE_SHUF checks pass the absolute 10^{-9} tolerance. The maximum persistence-diagram difference is zero for both H_0 and H_1; the largest difference among g_1–g_3 is 1.51×10^{-14}. Point-cloud H_1 measures loop geometry, not temporal return. Repeated three-seed calculations are bitwise identical; the original independent-projection seed 1337 still rounds to 0.0326 and 5.2649. The original and supplemental CPU runs take about 211 and 279 seconds, respectively, including their five-seed benchmarks and reporting.

Per-seed values, frozen thresholds, coefficients, scalers, predictions, bootstrap intervals, categories, shuffle checks, environment versions and original output hashes are recorded under `experiments/results/toy_v2_robustness/`, with supplemental outputs in `amendment_1/`. Degenerate bootstrap intervals such as $[1, 1]$ describe the sampled seeds under the fixed construction, not certainty about other settings. These remain synthetic pipeline checks; none of these results concerns LLM activations, user identity, or empirical return to an attractor.

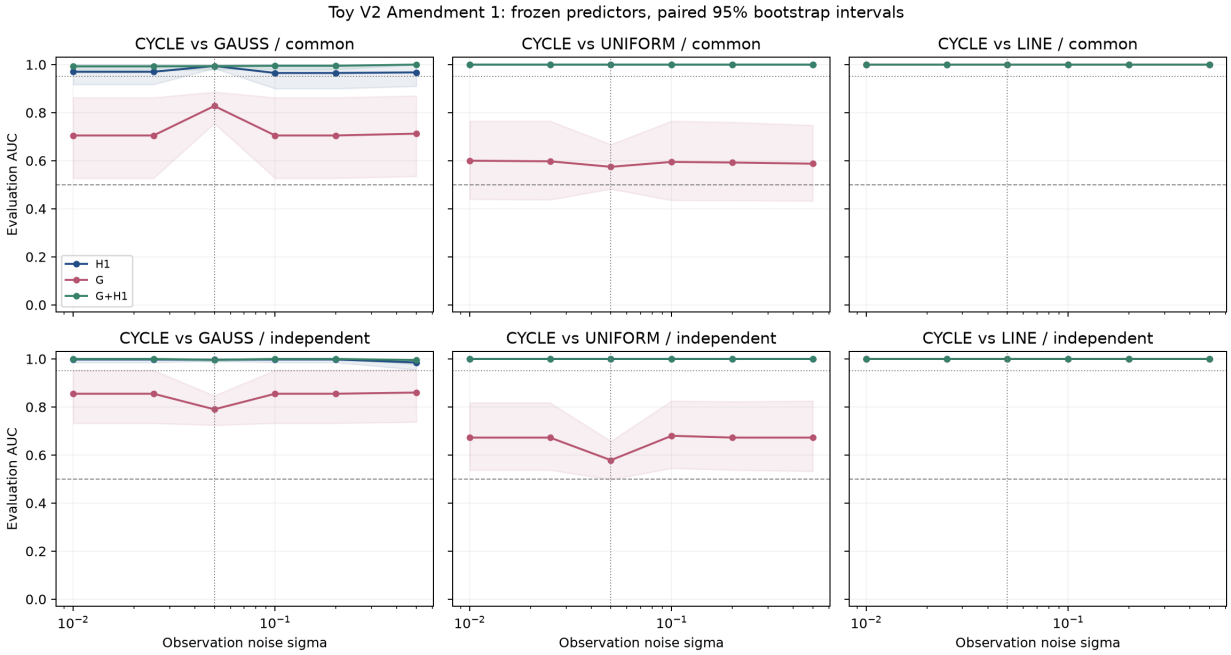


Figure 4: Amendment 1 noise sweep: evaluation AUCs of frozen predictors, with paired 95% bootstrap intervals. Top: common projection; bottom: independent projections. The primary 0.05 point uses 100 evaluation seeds per class (100–199); all other points use 20 (20–39). Changes across these points therefore also reflect different evaluation seed sets. No levels are pooled.

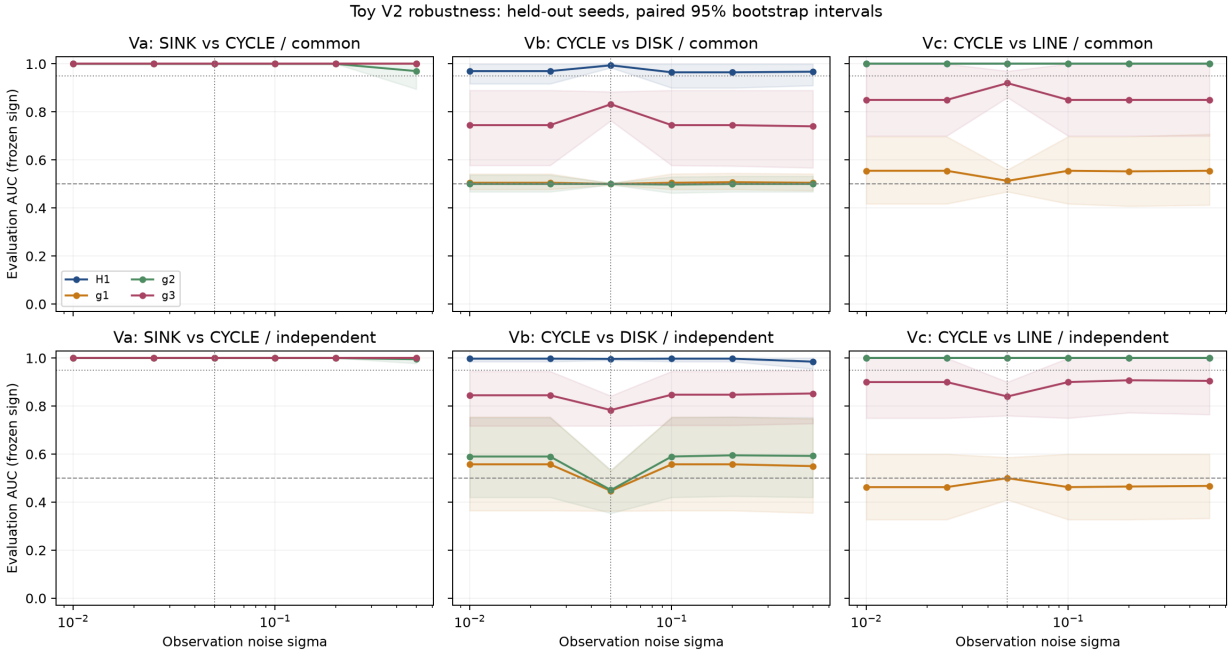


Figure 5: Original committed noise sweep, preserved unchanged. The primary noise point and other points use the different seed sets specified in Figure 4; all statistics use calibration-frozen signs.

Table 7: Original committed primary evaluation, retained verbatim numerically: Va: SINK versus CYCLE; Vb: CYCLE versus DISK (GAUSS); Vc: CYCLE versus LINE. AUC and accuracy with 95% bootstrap intervals; signs and thresholds were frozen using calibration seeds.

Comparison / control	Statistic	AUC (95% CI)	Accuracy (95% CI)
Va	H_1	1.0000 (1.0000, 1.0000)	0.9900 (0.9750, 1.0000)
Va	g_1	1.0000 (1.0000, 1.0000)	0.9950 (0.9850, 1.0000)
Va	g_2	1.0000 (1.0000, 1.0000)	0.9950 (0.9800, 1.0000)
Va	g_3	1.0000 (1.0000, 1.0000)	0.9950 (0.9800, 1.0000)
Va	$g_{1,\mathrm{half}}$	1.0000 (1.0000, 1.0000)	0.9950 (0.9850, 1.0000)
Vb	H_1	0.9942 (0.9842, 1.0000)	0.9850 (0.9650, 1.0000)
Vb	g_1	0.5000 (0.4962, 0.5031)	0.5000 (0.5000, 0.5000)
Vb	g_2	0.5003 (0.4968, 0.5045)	0.5000 (0.5000, 0.5000)
Vb	g_3	0.8321 (0.7637, 0.8843)	0.7000 (0.6449, 0.7550)
Vb	$g_{1,\mathrm{half}}$	0.7920 (0.7473, 0.8369)	0.7100 (0.6599, 0.7600)
Vc	H_1	1.0000 (1.0000, 1.0000)	1.0000 (1.0000, 1.0000)
Vc	g_1	0.5129 (0.4678, 0.5590)	0.5250 (0.4800, 0.5700)
Vc	g_2	1.0000 (1.0000, 1.0000)	0.9950 (0.9850, 1.0000)
Vc	g_3	0.9200 (0.8600, 0.9700)	0.9500 (0.9150, 0.9750)
Vc	$g_{1,\mathrm{half}}$	0.6381 (0.5961, 0.6859)	0.6350 (0.5900, 0.6850)

4.4 What the toy experiments do and do not establish

Taken together, the two toy experiments support a restricted interpretation of LEM.

Supported in the present paper.

- User-conditioned latent trajectories can be geometrically distinguishable above chance in a controlled synthetic separability regime.
- On the one prescribed sink/cycle pair, H_1 maximum lifetimes differ; their descriptive ratio is 161.6.
- The amended multi-seed Toy V2 check has category *incremental value* under the specified generators, fixed predictors and evaluation rule.
- The synthetic constructions provide concrete code paths and motivate preregisterable measurements for future empirical tests.

Not established in the present paper.

- That real frontier-model activation traces exhibit the same trajectory classes.
- That interaction signatures correspond to psychological or neuroscientific identity.
- That high-dimensional synthetic separability implies real-world user identifiability.
- That cross-version stability has already been demonstrated empirically.
- That Toy V2 generalizes beyond the specified synthetic generators, projections, noise levels and fixed predictors, or establishes universal superiority of persistent homology over geometric methods.
- That LEM currently enables deployment-grade personalization, safety, or forensic attribution.

5 Related Work

LEM is related to several existing lines of research, but does not coincide exactly with any single one of them. Its closest neighbors lie in user modeling, latent behavioral representations, mechanistic interpretability, and topological analysis of neural representations.

5.1 User modeling and latent behavioral structure

A first neighboring tradition concerns latent user representations and personalization. In these approaches, user-specific state is typically modeled through compressed embeddings, memory summaries, or preference vectors. LEM is compatible with this general objective, but differs in one central respect: it treats repeated user–model interaction not primarily as a static user representation problem, but as a trajectory-level object. Related work on latent state persistence in LLMs has raised important questions about whether models maintain stable internal representations across turns at all [9]. LEM does not presuppose that real LLMs do so; rather, it asks whether the structure of user-conditioned trajectories is detectable if such persistence were present.

A closer point of contact is recent work on persona vectors. Chen et al. identify directions in activation space that correspond to behavioral or character traits—such as sycophancy, hallucination

propensity, or harmfulness—and demonstrate that these directions are causally effective: injecting or suppressing them during inference predictably alters model behavior [6]. This provides direct empirical evidence that the latent spaces through which LEM posits user-conditioned trajectories contain structured, behaviorally meaningful geometry. LEM builds on this foundation but shifts the focus from static trait directions to the temporal dynamics of repeated interaction trajectories and the possibility of deriving geometric and topological signatures from them.

5.2 Mechanistic interpretability and feature geometry

LEM is also related to mechanistic interpretability work that studies the internal organization of model representations. In particular, dictionary-learning and sparse-autoencoder work argues that model activations can be decomposed into more interpretable features than individual neurons alone [4, 13]. These results matter for LEM because they provide evidence that model representations are structured enough to support nontrivial latent analyses. A related direction is the literature on attribution graphs and circuit tracing, which models internal feature interactions as structured computational pathways [1]. LEM differs in being trajectory-centric rather than feature- or circuit-centric.

5.3 Topological analysis of representation structure

A third neighboring tradition concerns topological data analysis in machine learning and language models. The foundational framework for applying persistent homology to data analysis was established by Carlsson [5]. More recently, Gardinazzi et al. apply zigzag persistence across successive layers of LLM activation spaces [8]. Their work does not establish a general advantage over linear probes for recognizing users. Tan et al. extend this direction to reasoning traces [12]. In a complementary line of work, Chia et al. probe latent subspaces of LLMs to identify adversarial states and demonstrate that structured regions in activation space can be isolated and manipulated [7]. This supports the broader premise that LLM latent spaces contain exploitable geometric and topological structure. LEM draws directly from this methodological tradition, especially in its use of persistent H_1 structure to distinguish topologically trivial from recurrent interaction regimes.

The narrower novelty claim of LEM is therefore not that topology can be applied to neural representations in general, but that user–LLM interaction can be modeled as a dynamical trajectory whose regime class may be summarized through geometric and topological signatures.

5.4 Dynamical systems and attractor perspectives

A fourth neighboring tradition treats language models through the lens of dynamical systems. Ramsauer et al. show that Transformer attention corresponds to the update rule of modern Hopfield networks, whose energy landscape admits multiple attractor types [11]. Bai et al. demonstrate that hidden layers of deep sequence models can converge to fixed points, formalizing the notion of equilibrium computation in neural networks [2]. Most directly relevant to LEM, Wang et al. provide empirical evidence that iterative text generation in LLMs converges to attractor cycles, including two-period limit cycles under successive paraphrasing [15]. These are observations for successive paraphrasing. Ko and Geiping report model-specific attractor-like behavior in multi-turn model–model conversations [10]; they do not establish a user-specific attractor in a human–model dialogue.

LEM differs from this line of work in three respects. First, it studies user-conditioned trajectories rather than generic model dynamics. Second, it operates in latent representation space rather than

in text space. Third, it employs topological analysis as a candidate way to summarize regime types. These are research aims, not experimentally established distinctions of this paper.

5.5 Positioning of the present contribution

The present paper should not be read as claiming to be the first connection between topology and language models, nor the first work on latent behavioral directions, nor the first dynamical-systems perspective on LLMs. Each of these components has established precedent. The narrower contribution of LEM is the synthesis: it introduces a single trajectory-based framework in which:

1. user-conditioned interaction is modeled dynamically (building on [2, 11, 15]);
2. attractor-like behavior becomes a first-class object of analysis, supported by evidence that latent spaces encode structured behavioral geometry [4, 6];
3. geometric identifiability and topological regime structure are treated jointly, using tools from persistent homology [5, 8].

LEM is therefore best understood as a proposed synthesis with a specific emphasis: the study of user–LLM interaction through trajectory dynamics, candidate attractor structure, and topological signatures. This review does not prove priority or uniqueness.

6 Discussion and Limitations

The main contribution of the present formulation is methodological. LEM is strongest when interpreted not as a completed explanatory theory, but as a compact research program that introduces a trajectory-based object of analysis and proposes geometric and topological tools for studying it.

The toy constructions provide two limited pipeline checks:

1. user-conditioned trajectories can be geometrically identifiable above chance in a controlled synthetic separability regime;
2. the one prescribed sink and one prescribed cycle produce different maximum H_1 lifetimes in the recorded example.

These observations do not establish that real language-model interaction traces behave in the same way. A future attractor test must specify the perturbation, recovery horizon, alternative null dynamics, and independent experimental unit before collecting confirmation data.

It bears emphasis that the perfect re-identification observed in Toy Experiment 1 is a consequence of the high-dimensional synthetic construction, not evidence for real-world user fingerprinting. In $d = 512$ dimensions, independently sampled random unit vectors are near-orthogonal by concentration-of-measure arguments, making separation trivially achievable in the absence of real-world confounds such as overlapping behavioral styles, model-induced compression, or adversarial noise. The value of Toy Experiment 1 lies not in the magnitude of the accuracy, but in confirming that the attractor-based update rule produces stable, structured trajectories in the controlled regime.

This work has several further limitations. First, all numerical illustrations use synthetic toy dynamics rather than internal activation traces from real language models; the new Toy V2 comparison is restricted to its pre-specified generators, matched controls and fixed predictors. Its original

result and later amendment are reported separately, since the original outputs already existed when the amendment arrived. Second, the notion of an interaction signature is operational and representational, not psychological or neuroscientific. Third, the current paper does not establish cross-version or cross-model stability. Fourth, no claim is made here that topological signatures are already useful for deployed personalization, safety, or forensic attribution. The later LEM-II-Light study on synthetic dialogues and real model activations did not establish a prespecified incremental benefit for its activation, time, or topology blocks. Its regime-classification design did not test user-specific attractor dynamics (see LEM-II-Light research note).

The paper deliberately avoids stronger claims about cognition, consciousness, or human identity. Its scope is restricted to a formal and computational framework for analyzing interaction dynamics.

7 Future Work

There are four immediate next steps:

1. collect independent quality-checked multi-turn traces from an open-weight model, with repeated profiles, held-out topics, and explicit text controls;
2. extend the synthetic comparison beyond the specified controls and fixed geometric predictor families, including diagram distances and temporal perturbation/return endpoints;
3. operationalize cross-version stability through diagram-based distances or related invariants;
4. connect LEM more explicitly to current interpretability workflows, including feature-based and persona-based latent analyses.

More ambitious directions, including cross-model user continuity, privacy implications, and robust trajectory-level defenses, should be treated as future hypotheses rather than present conclusions.

The framework documentation and code are available at `https://github.com/LAKITALKS/LEM`. The archival edition v2.1.0, originally labeled "bibliography-corrected" despite errors in the archived PDF (see `ARCHIVE_ERRATUM.md`), is citable at doi:10.5281/zenodo.21268033. That DOI identifies the archive, not this unpublished v2.3.0 working draft. The current published version is v2.2.0, DOI 10.5281/zenodo.22973258.

8 Conclusion

LEM provides a compact and testable framework for analyzing user–LLM interaction dynamics as trajectories rather than isolated states. Its central contribution is to combine attractor-based intuition with a topological reformulation that makes convergent and cyclic regimes measurable in principle. The current paper presents a first version of this program: a conceptual formulation and synthetic pipeline checks. The amended Toy V2 analysis is assigned *incremental value*, with *incremental value* for GAUSS and *clear advantage* for UNIFORM; the original committed rule gave *partial*. These categories refer only to this toy and these fixed comparisons. Point-cloud H_1 measures loop geometry, not temporal return. Whether these ideas hold for user-specific dynamics in model internals remains an open empirical question. However, even at the present stage, LEM provides a clearer basis for asking that question in a controlled and scientifically tractable way.

Research and tool disclosure

"Eudora" in the project name denotes AI collaboration, not a second human author. AI tools assisted with drafting, code, and review; the named author is responsible for the claims and released materials. Neither of the toy checks in this manuscript includes human participants.

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